EVST COURSE PROJECT 18 · TEAM 2 · FINAL EVALUATION

Environmental change before and after the First Pandemic

Fossil-pollen evidence for land use in Europe and the Mediterranean, 250–900 CE

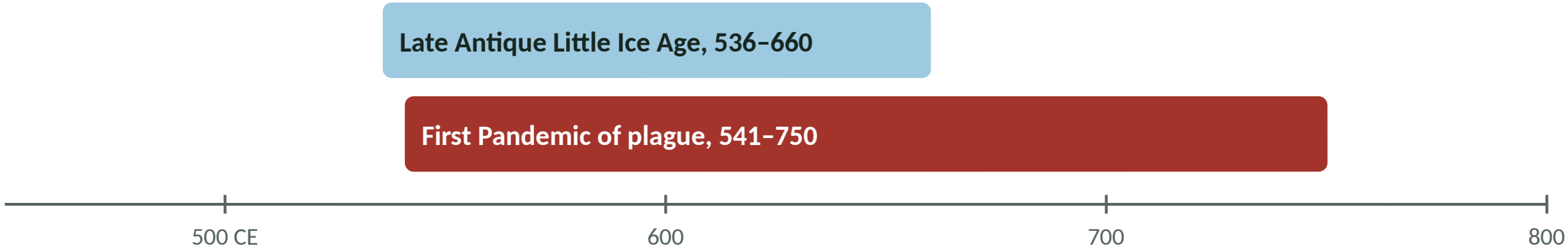
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IIIT Hyderabad · Monsoon 2026



THE QUESTION

Two shocks hit the late Roman world at the same time



If plague mortality emptied the countryside, farming pollen should fall and tree pollen rise after 541. But the cold arrived in the same years.

1. **Did land use change after 541?**
2. **Can a change be separated from the LALIA cooling?**
3. **Does the HYDE 3.2 land-use model agree with the pollen?**

METHOD · THE PROXY

Fossil pollen shows how the land around a lake was used



Cereal

Cerealia-type, Secale, Triticum, Hordeum, Avena

Arable fields



Pastoral

Plantago lanceolata, Rumex acetosa/acetosella, Rumex

Grazed meadows



Cultivated trees

Olea, Juglans, Castanea, Vitis (OJCV)

Olive, walnut, chestnut and vine groves



Natural trees

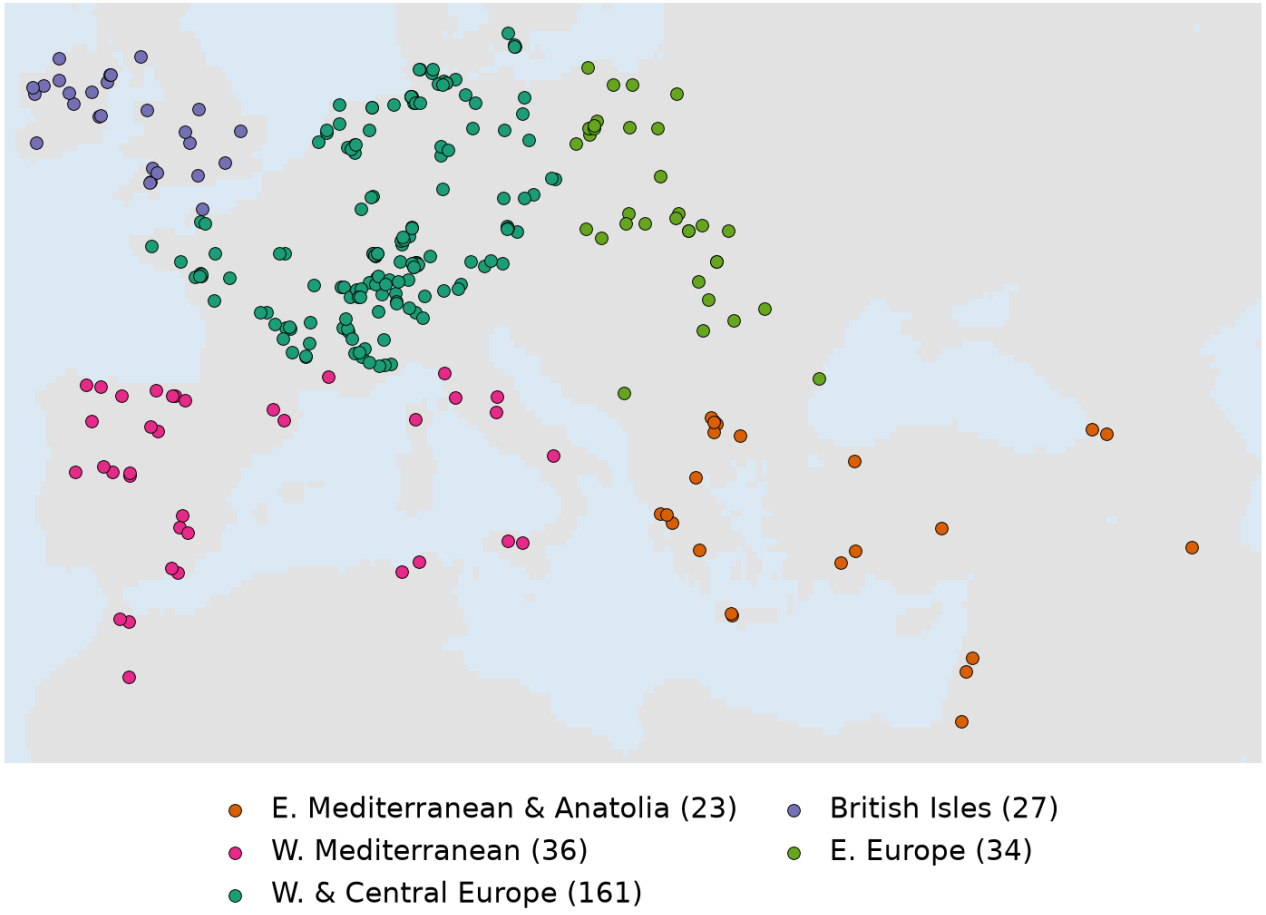
All tree and shrub pollen except OJCV

How much woodland remained

All indicators are % of the terrestrial pollen sum. “All anthropogenic” = cereal + pastoral + cultivated trees. We also measure rarefied taxon richness, E(S100). If plague emptied the fields: farming indicators fall, natural trees rise.

DATA

1,099 Neotoma datasets, 281 core sites



12,508

pollen samples dated 0–1200 CE

984 sites

from 1,064 records; 14 duplicate records removed, cores from one lake pooled

281

core sites: ≥ 2 samples both before (400–540) and after (541–750 CE)

23

core sites in the Eastern Mediterranean & Anatolia (13 at mid-term)

Compare before and after 541, and against what came before

Earlier change 250–399	Before 400–540	After: plague period 541–750	Following 751–900

Before vs after

Change in each indicator at each site; paired Wilcoxon test, bootstrap 95% CI

Placebo

Is the rate of change across 541 faster than in 250–399 → 400–540?

Window sensitivity

Narrower windows, boundary ±50 yr, first 60 years only, following period

Regression

Site fixed effects: land use ~ temperature + plague period + trend, 300–900 CE

Dating uncertainty

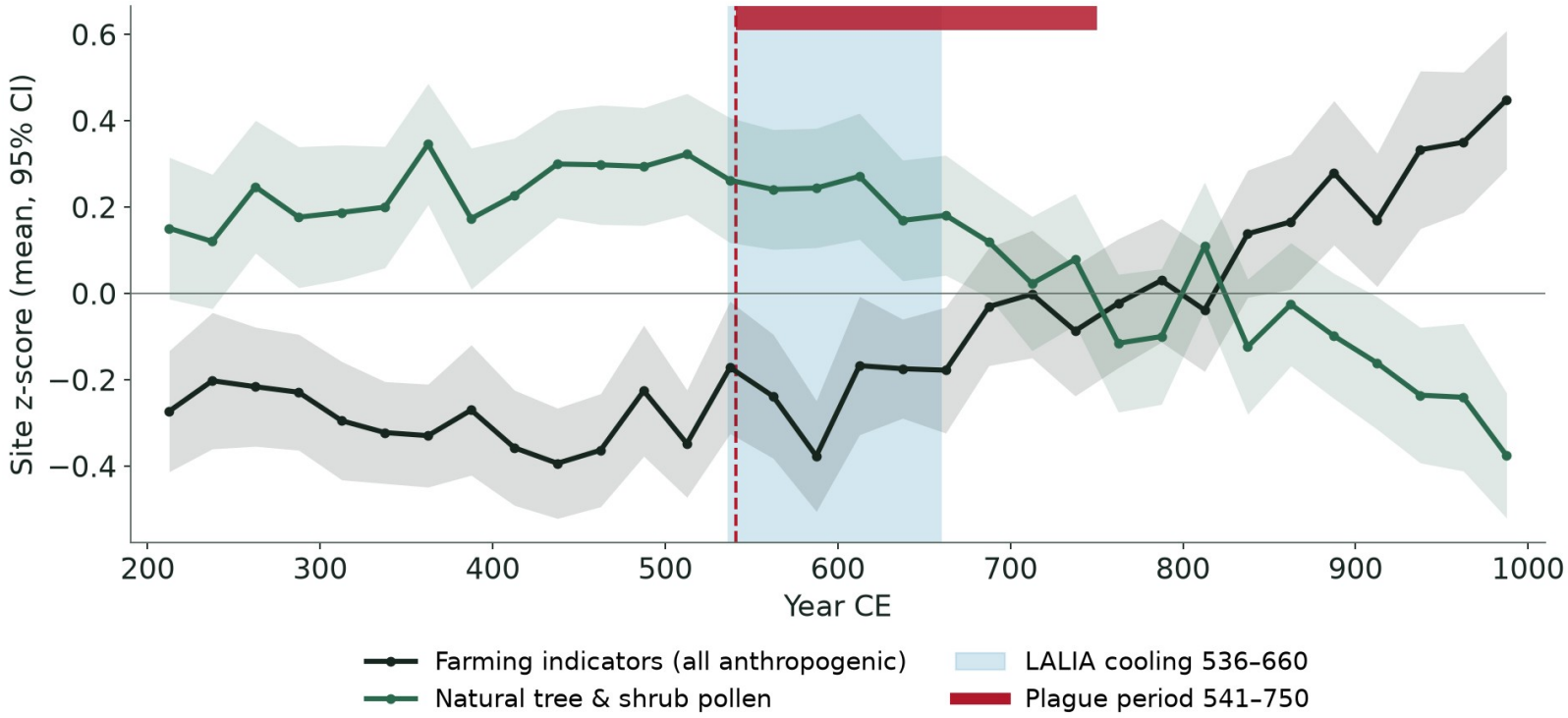
1,000 Monte Carlo realisations of every age model

HYDE 3.2

The model's cropland and grazing around the same sites, same windows

RESULT 1 · EUROPE-WIDE

Farming grew after 541; it did not collapse



+0.28 pp

all farming indicators, 541–750 vs 400–540 ($p < 0.001$)

−0.70 pp

natural tree & shrub pollen ($p = 0.004$)

56% vs 30%

of sites where cereal pollen rose vs fell

RESULT 1 · ROBUSTNESS

The rise holds for every window except the first 60 years

Test	Windows (CE)	Sites	All farming (pp)	p
Main comparison	400–540 → 541–750	281	+0.28	< 0.001
Narrower windows	450–540 → 541–640	135	+0.21	0.029
Boundary 50 yr earlier	350–490 → 491–700	288	+0.14	< 0.001
Boundary 50 yr later	450–590 → 591–800	288	+0.26	< 0.001
Plague period vs 751–900	541–750 → 751–900	302	+0.37	< 0.001
First 60 years only	**450–540 → 541–600**	**79**	**+0.05**	**0.40**

Placebo

+0.14 vs 0.00

pp per century, across 541 vs the period before (226 sites, p = 0.007)

In the first 60 years of plague alone (79 sites with enough samples) nothing changes significantly, in either direction.

RESULT 2 · REGIONS

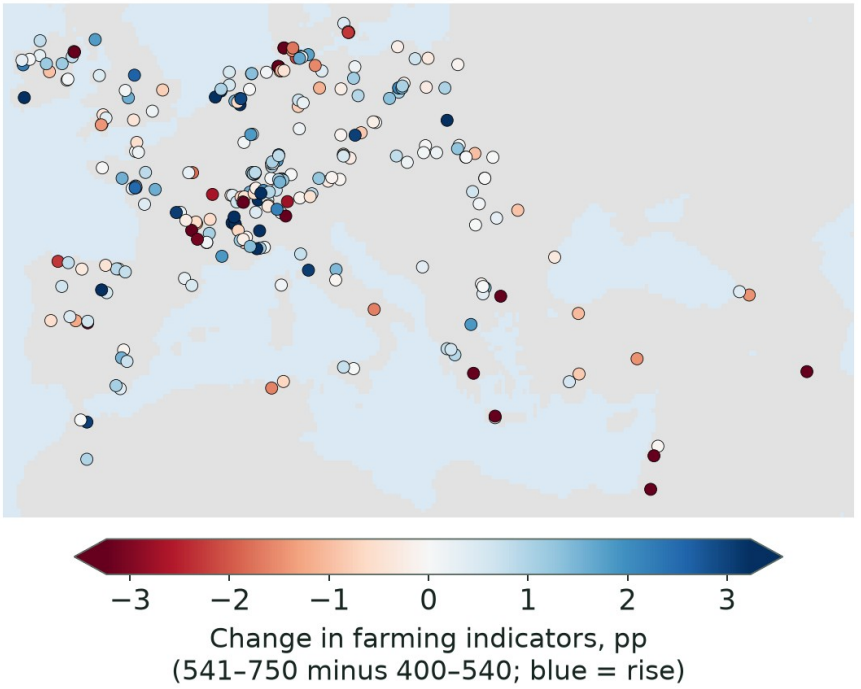
Western and central Europe carries the signal

Region	Sites	All farming (pp)	p	Dating test*
W. & Central Europe	161	+0.30	**< 0.001**	**100%**
W. Mediterranean	36	+0.50	0.057	19%
E. Europe	34	+0.14	**0.011**	14%
British Isles	27	+0.23	0.12	3%
E. Mediterranean & Anatolia	23	−0.08	0.18	24%

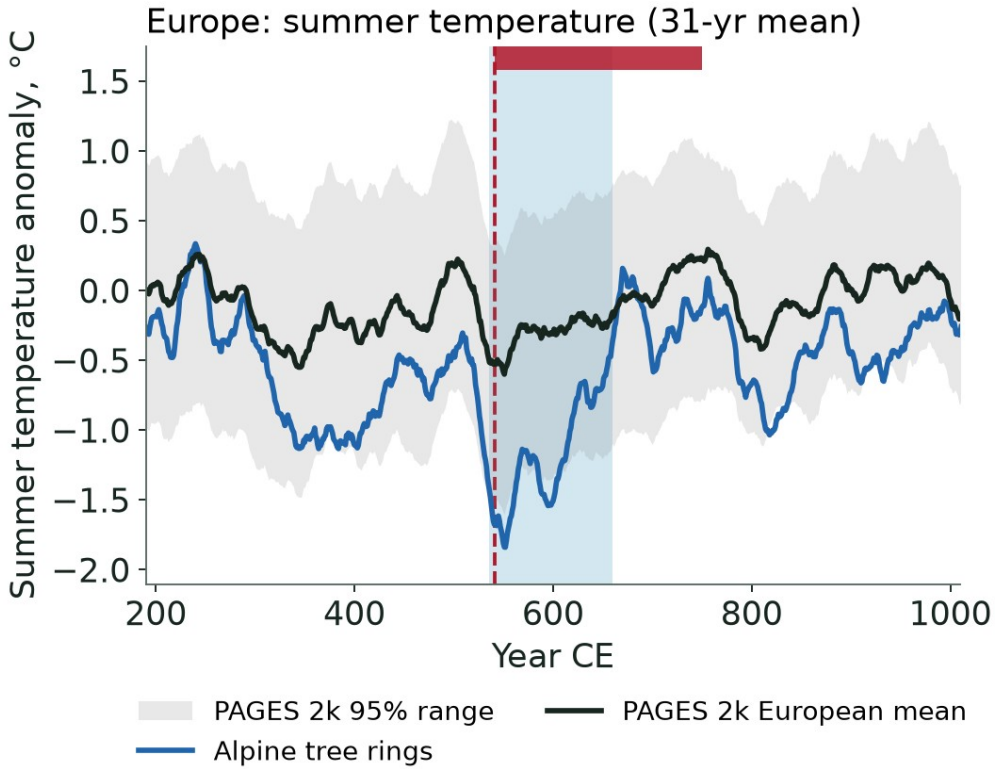
* share of 1,000 age-model realisations in which the change stays significant. Red p: < 0.05.

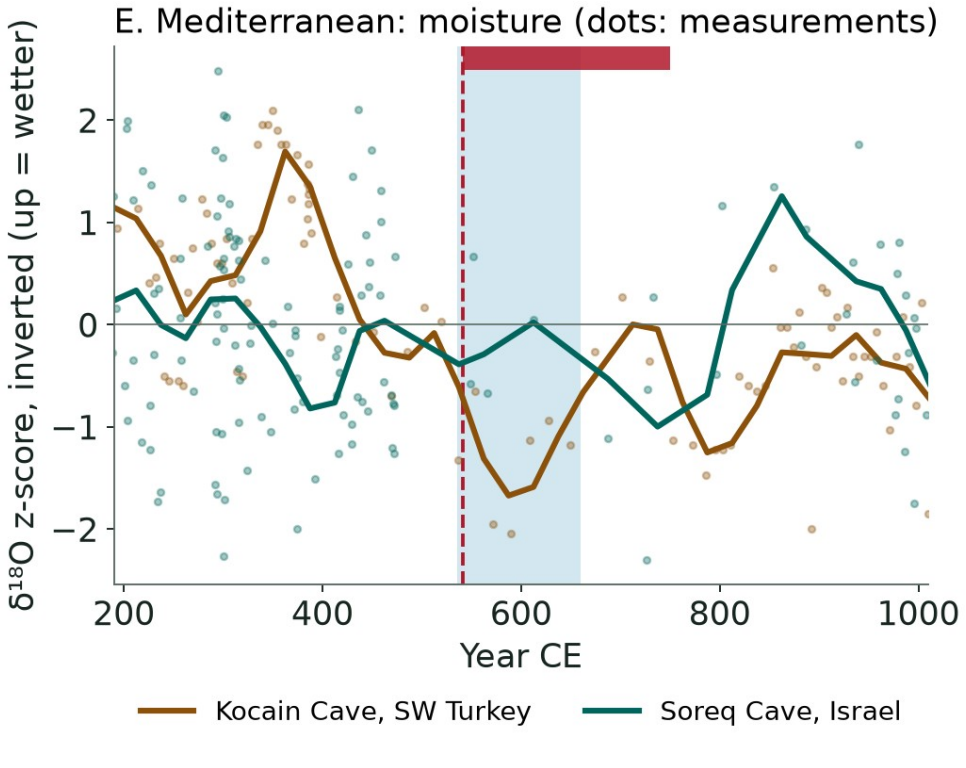
Eastern Europe also rises, matching Slavic settlement, but the result does not survive dating uncertainty.

Eastern Mediterranean: the only region leaning downward; not significant.



Europe cooled only slightly; the East was not one climate

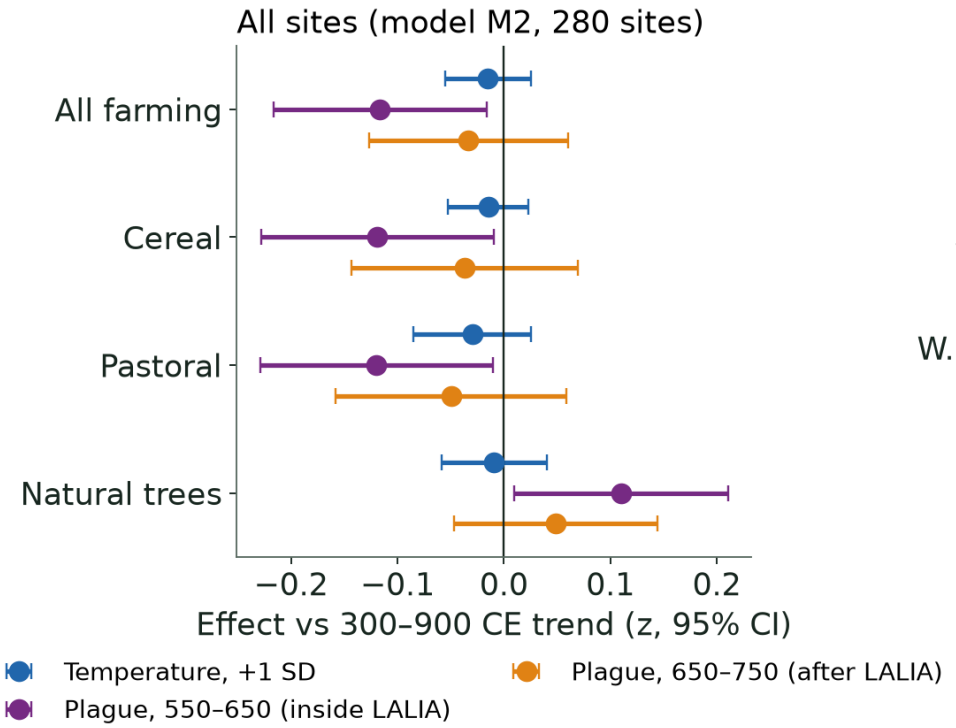


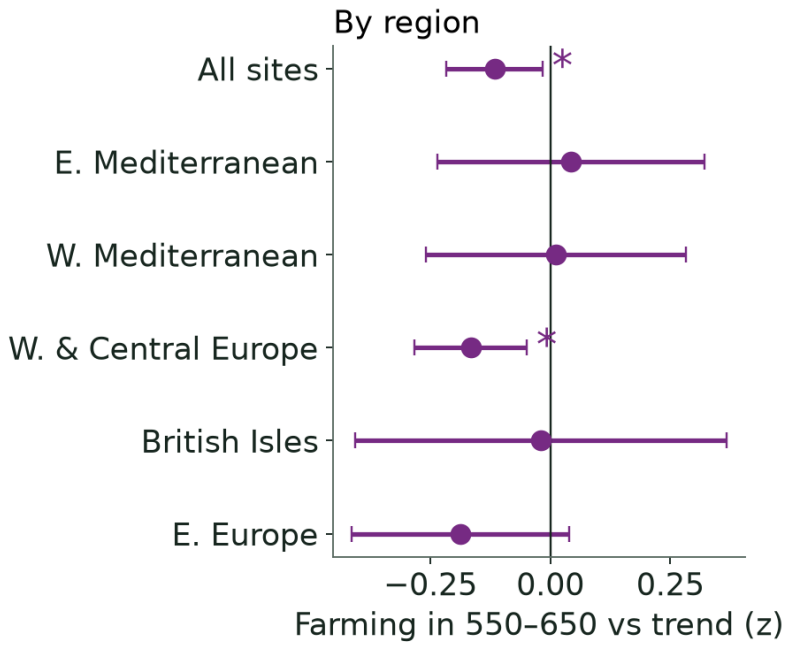


European mean cooled 0.25–0.32 °C (Alpine series 0.61 °C) from 400–535 to 536–660 CE; the reconstruction's 95% range is about ±0.9 °C. Kocain dried; Soreq did not, but has only 4 measurements in 541–660.

RESULT 4 · REGRESSION

No temperature effect; a short pause in 550–650





−0.005 SD

effect of +1 SD (0.22 °C) warmer summers (p = 0.78)

−0.12 SD

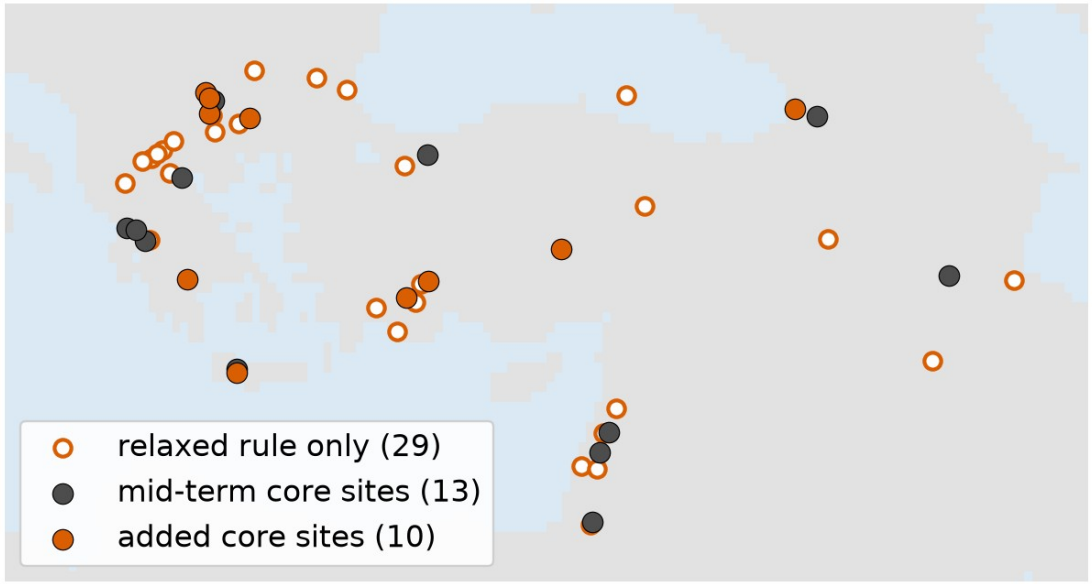
farming in 550–650 vs trend (p = 0.024); W. & Central Europe −0.17 SD

−0.03 SD

in 650–750, back on trend (p = 0.47)

With a curved (quadratic) trend the pause disappears (+0.01 SD, p = 0.76): a modest slowdown, not a collapse.

A bigger sample, still no significant decline



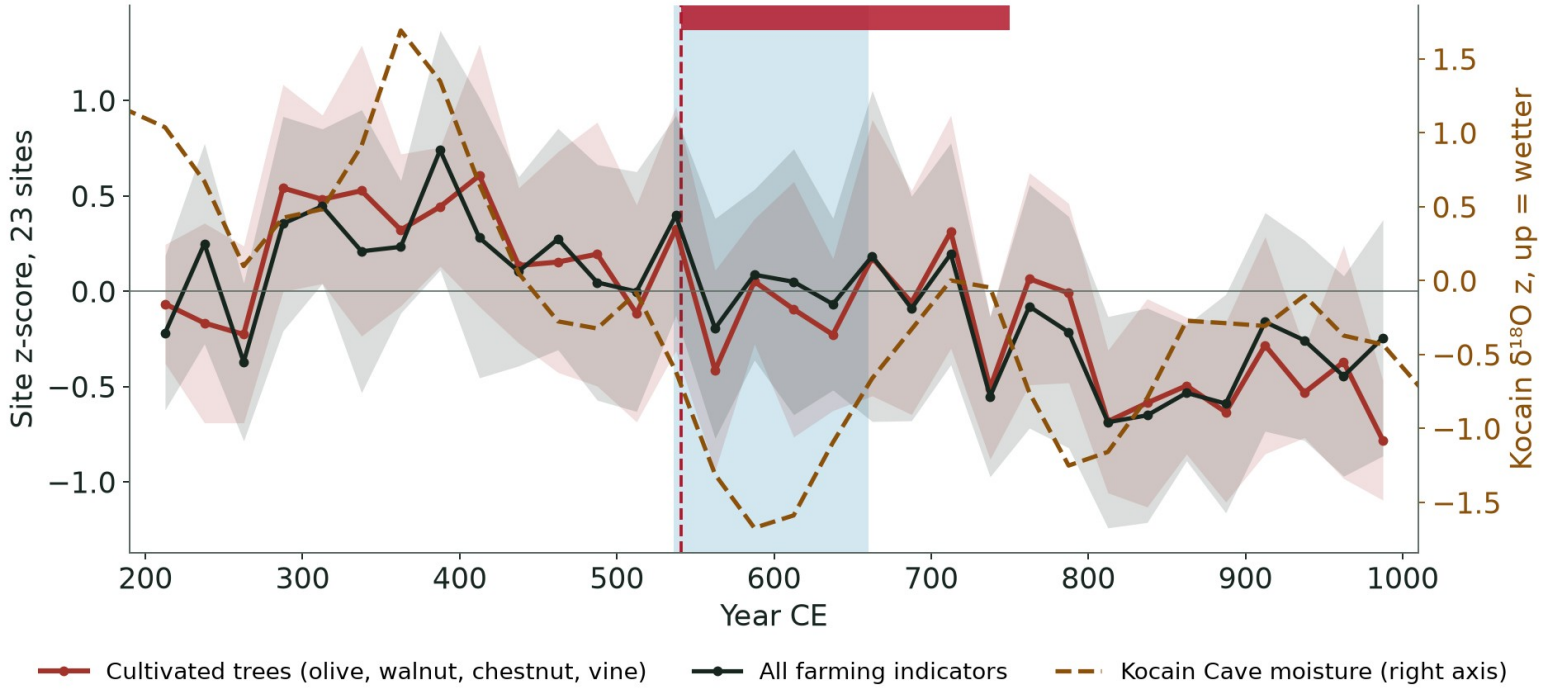
Sample	Cultivated trees (pp)	All farming (pp)
Mid-term (13 sites)	−0.42 (p = 0.15)	−0.06 (p = 0.38)
Final core (23 sites)	−0.07 (p = 0.12)	−0.08 (p = 0.18)
Relaxed rule (52 sites)	0.00 (p = 0.33)	−0.07 (p = 0.43)

What changed since mid-term: polygon-stored and calendar-dated records recovered, 23 core sites (13 before). The mid-term arboriculture decline shrinks from −0.42 to −0.07 points and is driven by Dead Sea, Kournas (Crete) and Lerna.

Region here = south of 43°N, east of 19°E: Greece, S. Bulgaria, Turkey, Levant, Georgia, NW Iran.

RESULT 5 · TIMING

Tree crops in the East were already lower before the plague



Peak 400–425 CE

cultivated-tree composite, then lower for the rest of the 5th and 6th centuries

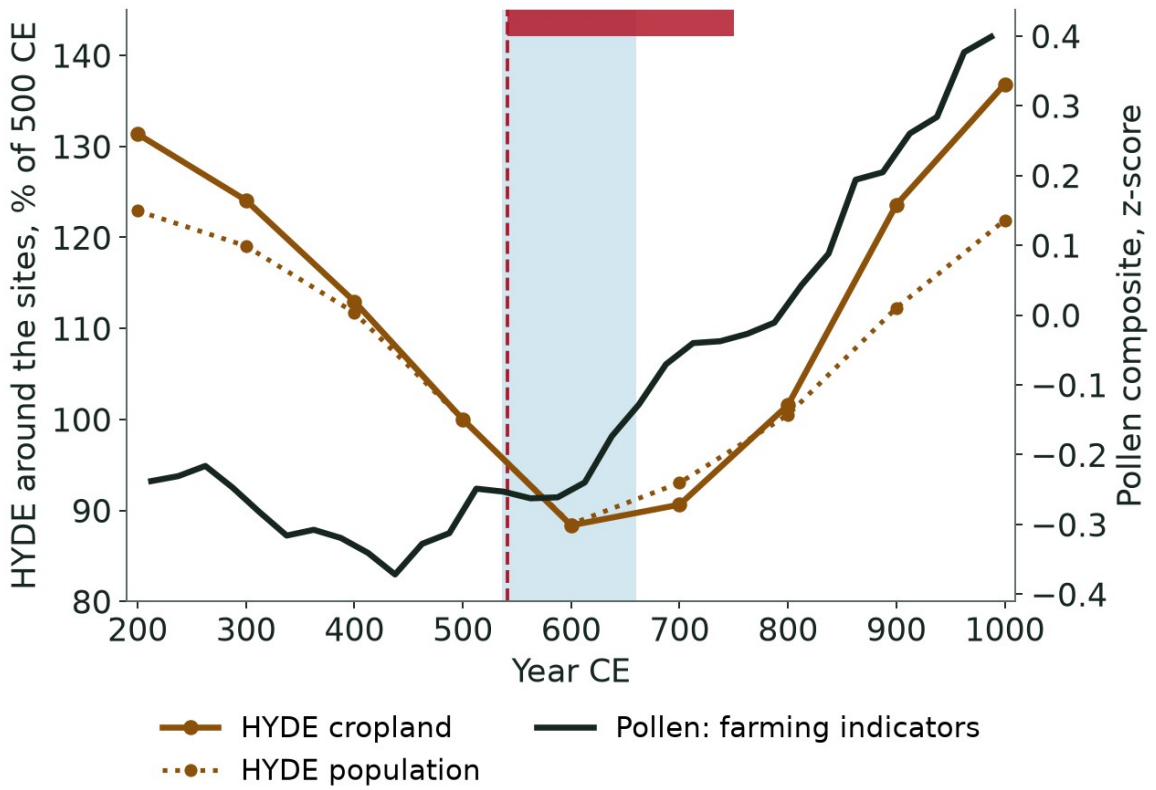
Kocain wettest 350–375 CE

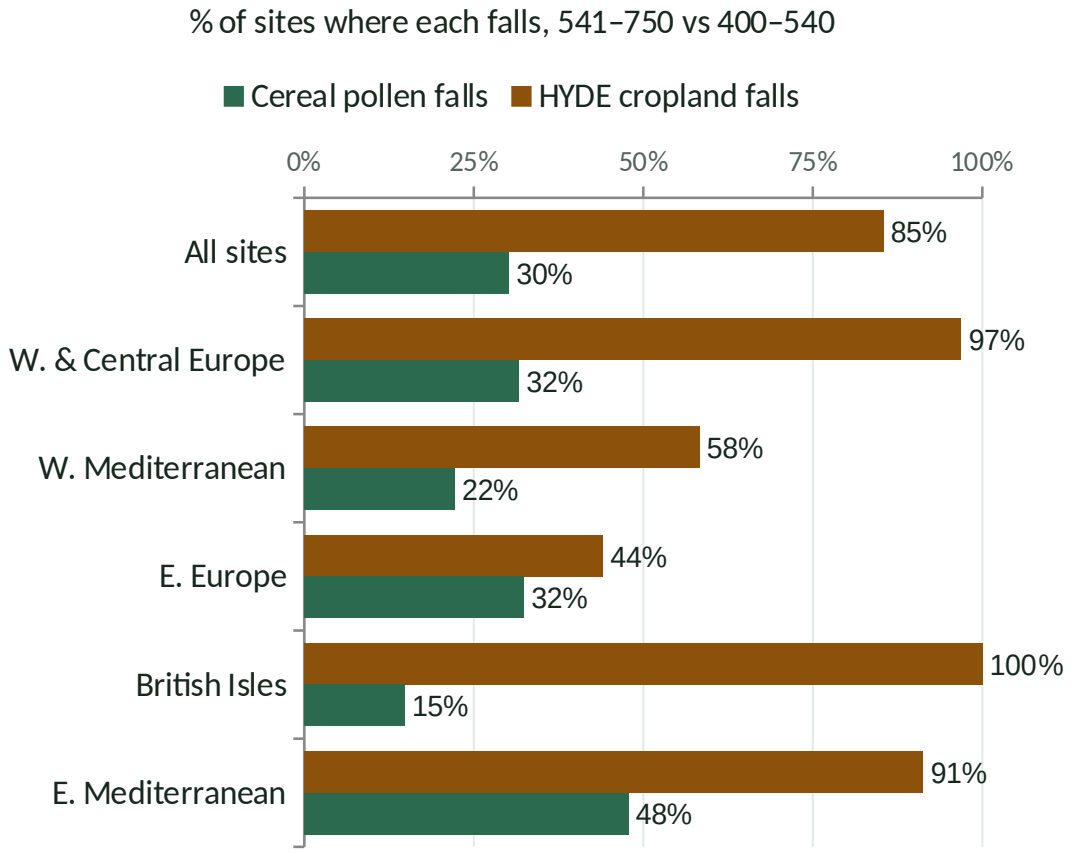
then drier: below its mean by 450–475 CE, driest 575–600 CE

No statistical link

no correlation after detrending, no moisture effect in the regression: drying is possible, not demonstrated

HYDE 3.2 cropland moves the wrong way





HYDE derives land use from population estimates, which dip to 88% of the 500 CE level in 600 CE. HYDE grazing, in contrast, rises at 70% of sites, like pastoral pollen. Spatial match with cereal pollen: ρ = 0.21.

RESULT 7 · DATING UNCERTAINTY

1,000 age-model runs: Europe holds, the regions do not

	Cereal	Pastoral	Cultivated trees	All anthropogenic	Natural trees	Richness
All sites	**↑ 100%**	↑ 90%	↑ 56%	**↑ 98%**	**↓ 95%**	**↑ 96%**
W. & Central Europe	**↑ 100%**	**↑ 93%**	↑ 74%	**↑ 100%**	**↓ 92%**	**↑ 98%**
W. Mediterranean	↑ 4%	↑ 11%	↑ 33%	↑ 19%	↓ 18%	↑ 12%
E. Europe	↑ 18%	↑ 11%	↑ 2%	↑ 14%	↓ 12%	↑ 6%
E. Mediterranean (23)	↓ 9%	↓ 4%	↓ 25%	↓ 24%	↑ 9%	↓ 16%
E. Med, relaxed (52)	↓ 12%	↓ 2%	↓ 15%	↓ 25%	↑ 7%	↓ 14%

Cells: share of realisations in which the change across 541 is significant ($p < 0.05$); arrow = direction. Ages moved within their reported errors (median σ 94 yr where none is reported), errors correlated along each core.

ANSWERS

What the pollen says

1 · CHANGE AFTER 541?

Yes: farming grew

Cereal, pastoral and all farming pollen rose at most sites and woodland fell, faster than before. No abandonment.

2 · SEPARATE FROM THE LALIA?

Partly

Temperature explains none of it. One modest pause, 550–650, overlaps both shocks and depends on how the trend is modelled.

3 · DOES HYDE AGREE?

No (cropland)

HYDE's population-driven cropland decline to 600 CE is not in the pollen; its map matches weakly. Its grazing rise does match.

SINCE THE MID-TERM

Everything planned is done, and one conclusion changed

- HYDE 3.2 compared with the pollen at every site
- Regional climate: PAGES 2k Euro-Med, Kocain and Soreq caves
- Fixed-effects regression: temperature vs plague period
- Eastern Mediterranean sample: 13 → 23 core sites
- 1,000-run Monte Carlo on dating errors
- 14 duplicate records removed; cores of one lake pooled

Revised

Mid-term: "a regional, not continental, mark, strongest in the eastern Mediterranean tree-crop economy."

Now: that decline is not significant with 23 sites, began around 425 CE and does not survive dating errors. We no longer support the regional claim.

Europe-wide rise: +0.20 → +0.28 pp, now with a fair placebo test.

LIMITATIONS

What this study cannot show

Pollen is relative and patchy Percentages, not areas; cereal pollen travels poorly; pollen records land use, not deaths.

Dating is coarse Median error ±176 years, as long as the windows; single plague waves cannot be resolved.

The plague period holds other events Lombard invasion, Avar and Slavic migrations, Persian–Byzantine war, Arab conquests.

Climate records are thin No regional temperature series for the 6th-century Mediterranean; European mean ±0.9 °C; the two caves disagree.

Unequal coverage 161 of 281 sites in W. & Central Europe; no open archaeological dataset covers the whole area.

CONCLUSION

The First Pandemic left no continental footprint of farmland abandonment in the pollen record

Farming kept expanding through the 6th–8th centuries, with a modest pause in 550–650 CE that temperature does not explain.

In the Eastern Mediterranean, tree crops were already lower a century before the plague.

HYDE 3.2 cropland, built on population estimates, moves opposite to the pollen.

Thank you · questions welcome

SOURCES AND REPRODUCIBILITY

Data, references and code

Data

- Neotoma Paleoecology Database (Williams et al. 2018): 1,099 pollen datasets
- HYDE 3.2.1 (Klein Goldewijk et al. 2017), doi:10.17026/dans-25g-gez3
- PAGES 2k Euro-Med (Luterbacher et al. 2016); Alpine tree rings (Büntgen et al. 2011)
- Kocain Cave (Jacobson et al. 2021); Soreq Cave (Orland et al. 2009), via NOAA

Key references

- Mordechai et al. 2019, PNAS: the Justinianic Plague, an inconsequential pandemic?
- Büntgen et al. 2016, Nature Geoscience: the Late Antique Little Ice Age
- Izdebski et al. 2022, Nature Ecology & Evolution: Black Death land use
- Kaplan et al. 2017, Land: historical land-use scenarios vs pollen
- Fuks et al. 2020, PNAS: Late Antique Negev viticulture



github.com/RiyasShaik/evst-project18-team2
./run.sh regenerates every number, table and figure in the report and on these slides (about 2 minutes, offline).